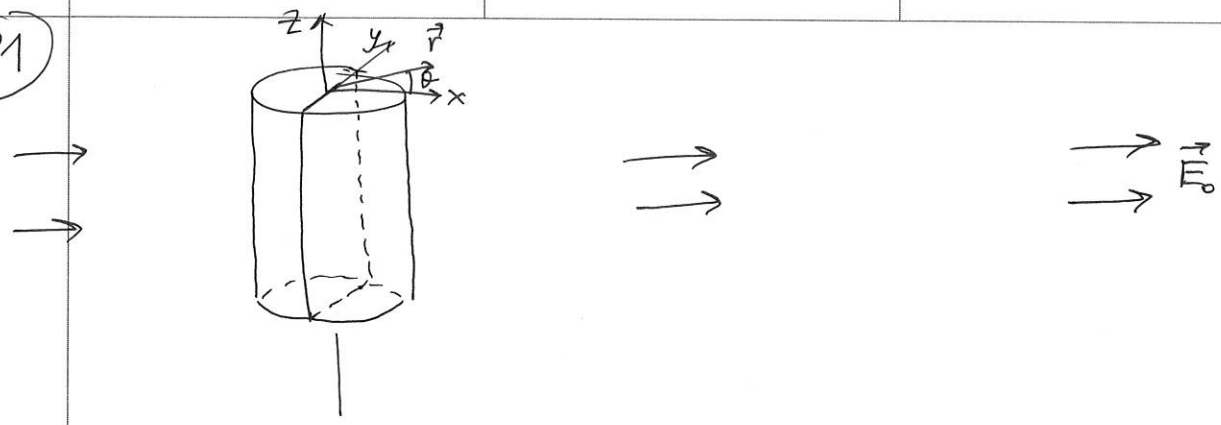
(P1)



a) $$\phi(r,\theta) = (a_0 + b_0\theta)\ln r + \sum_\nu \left(\alpha_\nu r^\nu + \beta_\nu r^{-\nu}\right)\left[a_\nu \cos(\nu\theta) + b_\nu \sin(\nu\theta)\right].$$

System is periodic in $\theta \to \theta + 2\pi \Rightarrow b_0 = 0$.

$\ln r$ diverges at $r = \infty \Rightarrow a_0 = 0$.

External field $\vec{E}_0$ is produced by $\phi_0 = -E_0 x = -E_0 r\cos\theta$

$\Rightarrow b_\nu = 0$.

Thus $\phi(r,\nu) = \sum_\nu \left(\alpha_\nu r^\nu + \beta_\nu r^{-\nu}\right)\cos(\nu\theta)$

and only $\nu = 1$ is present: $\phi = \left(\alpha r + \frac{\beta}{r}\right)\cos\theta$

For $r \to \infty$ $\phi = \alpha r\cos\theta = -E_0 r\cos\theta \Rightarrow \alpha = -E_0$.

For $r = R$ $\phi = 0 \Rightarrow -E_0 R + \frac{\beta}{R} = 0 \Rightarrow \beta = E_0 R^2$.

Thus $\phi(r,\theta) = -E_0 r\cos\theta + \frac{E_0 R^2\cos\theta}{r}$

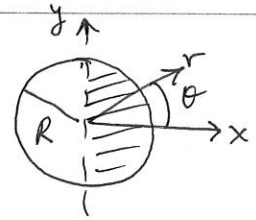
In cartesian coordinates $\phi(x,y) = -E_0 x + \frac{E_0 R^2 x}{x^2+y^2}$

$\vec{E} = -\vec{\nabla}\phi = (E_x, E_y)$

$$\begin{cases} E_x = E_0 - \frac{E_0 R^2}{x^2+y^2} + \frac{E_0 R^2 x \cdot 2x}{(x^2+y^2)^2} = E_0 + \frac{E_0 R^2 (x^2-y^2)}{(x^2+y^2)^2} \\ E_y = \frac{E_0 R^2\, 2xy}{(x^2+y^2)^2} \end{cases}$$

In polar coordinates: $E_x = E_0 + \frac{E_0 R^2\cos(2\theta)}{r^2}$, $E_y = \frac{E_0 R^2 \sin(2\theta)}{r^2}$.

b)



To calculate force use stress tensor

$$\sigma_{ij} = \varepsilon_0 \left(E_i E_j - \frac{1}{2} \delta_{ij} \vec{E}^2 \right)$$

$$\sigma_{ij} = \varepsilon_0 \begin{pmatrix} E_x^2 - \frac{1}{2}\vec{E}^2 & E_x E_y & 0 \\ E_x E_y & E_y^2 - \frac{1}{2}\vec{E}^2 & 0 \\ 0 & 0 & -\frac{1}{2}\vec{E}^2 \end{pmatrix} = \varepsilon_0 \begin{pmatrix} \frac{1}{2}(E_x^2 - E_y^2) & E_x E_y & 0 \\ E_x E_y & \frac{1}{2}(E_y^2 - E_x^2) & 0 \\ 0 & 0 & -\frac{1}{2}(E_x^2 + E_y^2) \end{pmatrix}$$

Clearly the force actors on <u>right-half</u> is along $\hat{x}$ axis.

$$F_x = \int_S \sigma_{xx}\, da_x + \sigma_{xy}\, da_y$$

where σ_{ij} is taken at $r = R$, $\theta \in (-\pi/2, \pi/2)$.
Part of the surface S passing along the diameter of the cylinder (inside the cylinder) does not contribute to F_x because electric field is zero there.

For $r = R$: $E_x = E_0 (1 + \cos 2\theta)$, $E_y = E_0 \sin 2\theta$.

Element of the area of cylinder: $da = l R d\theta$
$\Rightarrow da_x = da \cos\theta$, $da_y = da \sin\theta$.

$$\sigma_{xx} = \frac{\varepsilon_0}{2}(E_x^2 - E_y^2) = \frac{\varepsilon_0}{2} E_0^2 \left[(1 + \cos 2\theta)^2 - \sin^2(2\theta) \right] =$$

$$= \frac{1}{2} \varepsilon_0 E_0^2 \left(1 + 2\cos 2\theta + \cos^2 2\theta - \sin^2 2\theta \right) = \frac{1}{2} \varepsilon_0 E_0^2 \left(1 + 2\cos 2\theta + \cos 4\theta \right).$$

$$\sigma_{xy} = \varepsilon_0 E_0^2 (1 + \cos 2\theta) \sin 2\theta.$$

$$F_x = \varepsilon_0 E_0^2 \ell R \left\{ \int_{-\pi/2}^{\pi/2} d\theta \cos\theta \ \frac{1}{2}\left(1+2\cos 2\theta + \cos 4\theta\right) \right.$$

$$\left. + \int_{-\pi/2}^{\pi/2} d\theta \sin\theta \left(1+\cos 2\theta\right) \sin 2\theta \right\} = \varepsilon_0 E_0^2 \ell R \left(\frac{8}{5} + \frac{16}{15}\right) =$$

$$= \frac{8}{3} \varepsilon_0 E_0^2 R \ell .$$

Force per unit length $\dfrac{F_x}{\ell} = \dfrac{8}{3} \varepsilon_0 E_0^2 R.$

The integral can be done like this:

$$\int_{-1}^{1} d(\sin\theta) \ \frac{1}{2}\left(1 + 2\left(1 - 2\sin^2\theta\right) + 1 - 2\sin^2(2\theta)\right)$$

$\sin 2\theta = 2\sin\theta \cos\theta$

$$+ 2\int_{-\pi/2}^{\pi/2} \underline{d\theta} \ \sin^2\theta \ \underline{\cos\theta} \left(1 + 1 - 2\sin^2\theta\right) =$$

$$= \int_{-1}^{1} dx \ \frac{1}{2}\left(1 + 2\left(1-2x^2\right) + 1 - 2\cdot 4x^2\left(1-x^2\right)\right)$$

$$+ \int_{-1}^{1} dx \ 2x^2\left(2-2x^2\right) = \frac{1}{2}\int_{-1}^{1} dx \left(4 - 4x^2 - 8x^2 + 8x^4\right)$$

$$+ 4\int_{-1}^{1} dx \ x^2\left(1-x^2\right) = 2\int_{-1}^{1} dx \left(1 - 3x^2 + 2x^4\right)$$

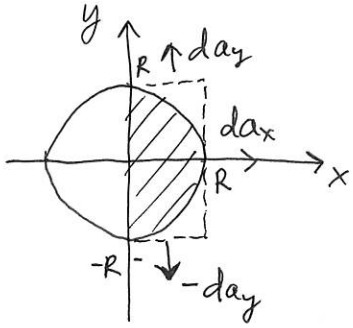
$$+ 4\int_{-1}^{1} dx \left(x^2 - x^4\right) = 4\left(x - x^3 + \frac{2}{5}x^5\right)\Big|_0^1$$

$$+ 4 \cdot 2\left(\frac{1}{3}x^3 - \frac{1}{5}x^5\right)\Big|_0^1 =$$

$$= 4\left(1 - 1 + \frac{2}{5}\right) + 8\left(\frac{1}{3} - \frac{1}{5}\right) = \frac{8}{5} + \frac{8}{15}\cdot 2 = \frac{8}{5} + \frac{16}{15} =$$

$$= \frac{24+16}{15} = \frac{40}{15} = \frac{8}{3}.$$

Another way to calculate F_x in problem 1b of Midterm 2.



$$F_x = \int \sigma_{xx}\, da_x + \int \sigma_{xy}\, da_y$$

$$\sigma_{xx} = \frac{\varepsilon_0}{2}\left(E_x^2 - E_y^2\right)$$

$$\sigma_{xy} = \varepsilon_0 E_x E_y$$

Instead of a semi-circle surface take rectangular (dashed line):

side: $da_x = l\, dy$ top: $da_y = l\, dx$ bottom: $da_y = -l\, dx$

$$E_x = E_0\left[1 + \frac{R^2(x^2-y^2)}{(x^2+y^2)^2}\right], \quad E_y = E_0\, \frac{R^2\, 2xy}{(x^2+y^2)^2}$$

$$F_x = E_0^2 l \left\{ \int_{-R}^{R} dy\, \frac{\varepsilon_0}{2}\left(E_x^2 - E_y^2\right)\Big|_{x=R} + \varepsilon_0 \int_0^R dx\, E_x E_y \Big|_{y=+R} \right.$$

$$\left. + \varepsilon_0 \int_0^R dx\, E_x E_y \Big|_{y=-R} \right\} = \varepsilon_0 E_0^2 l \cdot$$

$$\cdot \left\{ \frac{1}{2}\cdot 2 \int_0^R dy \left(\left[1 + \frac{R^2(R^2-y^2)}{(R^2+y^2)^2}\right]^2 - \left[\frac{2R^3 y}{(R^2+y^2)^2}\right]^2 \right) \right.$$

$$+ \int_0^R dx \left[1 + \frac{R^2(x^2-R^2)}{(x^2+R^2)^2}\right]\left(\frac{+2R^3 x}{(x^2+R^2)^2}\right)$$

$$\left. + \int_0^R dx \left[1 + \frac{R^2(x^2-R^2)}{(x^2+R^2)^2}\right]\left(\frac{-2R^3 x}{(x^2+R^2)^2}\right) \right\}$$

Now let $y = Ru$, $x = Rv$ in the two integrals correspondingly.

$$F_x = \varepsilon_0 E_0^2 \ell R \left\{ \int_0^1 du \left[\left(1 + \frac{1-u^2}{(1+u^2)^2}\right)^2 - \frac{(2u)^2}{(1+u^2)^4} \right] \right.$$

$$\left. + 2 \int_0^1 dv \left[1 + \frac{v^2-1}{(v^2+1)^2} \right] \frac{2v}{(v^2+1)^2} \right\} = \frac{8}{3} \varepsilon_0 E_0^2 R \ell.$$

The integral reduces to

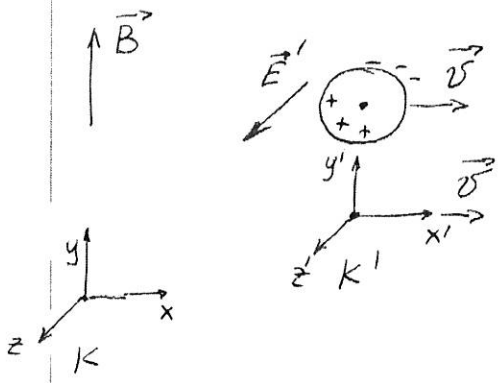
$$\int_0^1 du \; \frac{4 + 12u^3 + 5u^4 + 4u^5 + 2u^6 + u^8}{(1+u^2)^4} = \frac{8}{3}.$$

The point is that to find the force acting on a body with the help of Maxwell's stress tensor one is free to choose <u>any</u> surface enclosing the body in question.

Since we found $\vec{E}(x,y)$ in terms of cartesian coordinates, it is then very natural to use the rectangular shape instead of a cylindrical one.

And the result is of course exactly the same.

Problem 2. Consider a reference frame K' moving with the sphere.



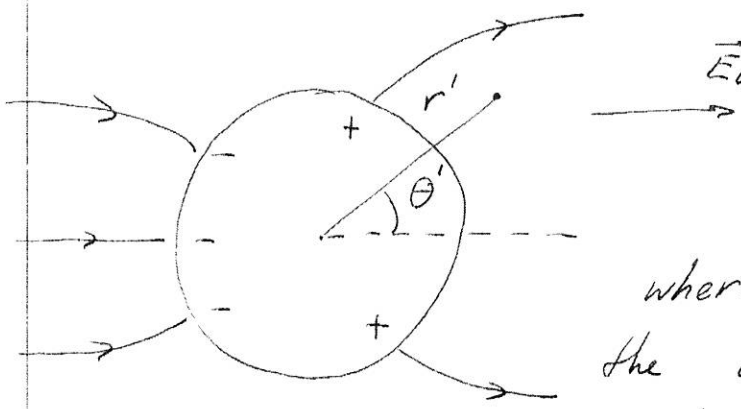
In that frame there is electric field (see Eq. 1.219)

$$E_z' = \gamma v B_y = v B_0$$

(to the lowest order in v/c)

In ref. fr. K' determination of the charge density is found from the Laplace equation

$$\nabla^2 \varphi = 0 \quad \text{with} \quad \varphi = \text{const}\,\Big|_{r=R} \quad \text{Boundary condition}$$



$$\vec{E}_0 = v B_0 \hat{z}$$

$$\varphi'(r', \theta') = f(r') \cos\theta'$$

where $f(r')$ satisfies the equation

$$\frac{1}{r'^2}\frac{d}{dr'}\left(r'^2 \frac{df}{dr'}\right) - \frac{2}{r'^2} f = 0$$

for $f(r) = r'^{\lambda}$ we get

$$\lambda(\lambda+1) - 2 = 0 \implies \lambda = 1, -2$$

The solution is thus

$$\varphi(r, \theta) = \left(a r' + \frac{b}{r'^2}\right)\cos\theta'$$

1) at $r' \to \infty$: $a = -E_0$ (only the external field "survives")
2) from the equipotentiality of the $r = R$ surface: $b = E_0 R^3$

All together

$$\varphi'(r',\theta') = v B_0 \left(-r' + \frac{R^3}{r'^2}\right)\cos\theta'$$

Surface charges,

$$\sigma'(\theta') = -\varepsilon_0 \frac{\partial \varphi}{\partial r'}\Big|_{r=R} = 3\varepsilon_0 v B_0 \cos\theta'$$

Transformation of charge density back to K could be performed using that $j^\alpha = (\rho c, \vec{j})$ is a 4-vector:

$$\rho c = \frac{c\rho' + \frac{v}{c} j_x'}{\sqrt{1 - v^2/c^2}}$$

Since in K' $\vec{j}' = 0$ we get to the lowest order in v/c simply

$$\rho = \rho' \implies \sigma = \sigma'$$

Since transformation of the angle θ is also $\theta = \theta'$ to the same order in v/c the final answer is

$$\sigma(\theta) = 3\varepsilon_0 v B_0 \cos\theta$$

Namely: $\cos\theta' = \dfrac{\cos\theta - \beta}{1 - \beta\cos\theta} \simeq (\cos\theta - \beta)(1 + \beta\cos\theta) = \cos\theta + \beta\cos^2\theta - \beta - \beta^2\cos\theta =$
$= (1-\beta^2)\cos\theta - \beta\sin^2\theta.)$

Therefore $\sigma(\theta) \approx 3\varepsilon_0 v B_0 \cos\theta$. Corrections to this result will produce $\sigma(\theta) = 3\varepsilon_0 v B_0 \cos\theta + O(v^2/c^2)$ terms.